

FEE3411: Control Systems A
Week 1
Part B — Control System Components (Abridged)

A CONCEPTUAL SURVEY

Reading

A shorter route through Part B of the Week 1 notes: the same devices, the same diagrams and the same conclusions, but the transfer functions are **stated** rather than derived, and explained in words rather than algebra. It does not replace `week-01-notes.pdf` — each section carries a *Full treatment* line naming exactly where the derivation lives.

The syllabus outcome here is ELO 1, “*describe the working principles*”, so describing is the requirement; the derivations belong to Week 2 and to FEE3412. Nothing examinable in FEE3411 is lost by reading this version — but do not skim §8, which is the idea the rest of the unit is built on.

By the end of this week you should be able to

- Say what each standard control component does, how it does it, and roughly what its transfer function looks like.
- Choose sensibly between an electrical, hydraulic and pneumatic solution for a given job, and defend the choice.
- Explain *in words* why almost every positional actuator has the transfer function $K/[s(\tau s + 1)]$, and why that single fact shapes Weeks 5, 6, 8 and 10.

1 Where the components sit

Everything in this document is one of four boxes in a loop you have already seen. Keep Figure 1 in view: when a later week labels a block “actuator”, this is what is physically inside it.

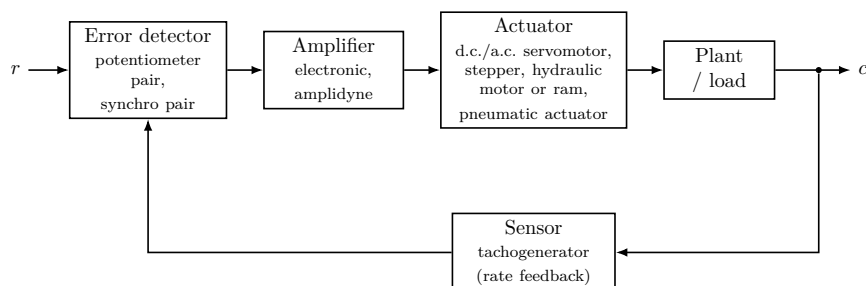


Figure 1: Where each family of components sits in the loop. Everything in Part B is one of these four boxes.

Key idea

There are only four jobs to do in a control loop: **measure the error, amplify it, act on the plant, and measure the output**. Every device below is a way of doing one of those four things. When you meet an unfamiliar component, the first question is which of the four it is.

2 Error detectors

The error detector performs the subtraction $r - c$ *physically*. Two standard devices do it for angular position.

The potentiometer pair

Two potentiometers share a d.c. supply; one is turned by the reference shaft, the other by the load shaft. The voltage between the two wipers is proportional to the difference between the shaft angles:

$$v_e = K_p (r - c), \quad [K_p] = \text{V rad}^{-1}.$$

The subtraction is done *by the wiring*, not by a circuit. This is the physical realisation of the summing junction you draw as $\otimes$.

For: cheap, d.c., and a *pure gain* with no dynamics. **Against:** the wiper is a sliding contact — friction, wear, noise, and a resolution set by the winding pitch.

The synchro pair

Where the shaft turns continuously, through many revolutions, or sits in a hostile environment, a sliding contact is unacceptable. The **synchro** (trade names *selsyn*, *autosyn*) replaces it with transformer action — no contact anywhere in the signal path.

Key idea

A synchro transmitter is a **transformer whose coupling depends on shaft angle**. Its rotor coil is the primary; its three stator coils are the secondaries; the voltage induced in each depends on the angle between that coil and the rotor. Input a shaft angle, get out three a.c. voltages that encode it.

Wire those three stator leads to a second, similar machine — a **control transformer**, built with a cylindrical rotor so its output impedance does not change as it turns — and the flux pattern of the first is reproduced inside the second. The voltage at the control transformer's rotor then depends on the angle *between the two rotors*. For small differences it is proportional to that difference:

$$e(t) \approx K_s (\theta - \alpha) \sin \omega_c t, \quad [K_s] = \text{V rad}^{-1}.$$

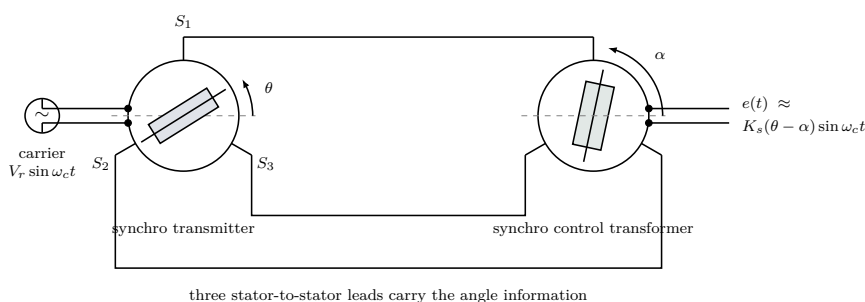


Figure 2: Synchro transmitter and control transformer used as an error detector. Cf. Nagrath & Gopal Fig. 4.12, p. 94. The three stator-to-stator leads carry the angle information; there is no sliding contact in the signal path.

The output is a carrier $\sin \omega_c t$ whose *amplitude* carries the error and whose *phase flips* when the error changes sign. This is **suppressed-carrier modulation**, and a loop whose internal signals look like this is called a **carrier control system**.

Common pitfall

In a carrier system the information is in the **envelope**, not the instantaneous value. A student who reads the error straight off an oscilloscope trace of $e(t)$, rather than its envelope, will conclude that the error is oscillating at 400 Hz while the shaft is standing perfectly still.

Full treatment: *main notes §5.2 (the coil voltages, the derivation of $e(t)$ from the rotor geometry, and where the small-angle approximation enters). Nagrath & Gopal 2nd ed. §4.3, p. 92.*

3 Actuators

The d.c. servomotor

The workhorse — a d.c. motor built for low inertia and rapid reversal. **Armature control** (field constant, armature voltage varied) is used almost universally; **field control** needs a cheaper amplifier but is slower, because the field winding is highly inductive. Driving inertia and friction, the armature-controlled motor has

$$\frac{\theta_m(s)}{V_a(s)} = \frac{K_m}{s(\tau_m s + 1)}.$$

Read that shape now and remember it — §8 is about why it keeps coming back. In words: *a constant voltage gives a constant speed* (hence the integrator, since position is the accumulation of speed), *and the motor takes time to reach that speed* (hence the first-order lag, because inertia and friction resist the change).

Large d.c. systems historically used an **amplidyne**, a rotating cross-field power amplifier. Obsolete — displaced by thyristor and IGBT drives — but the name appears in older textbook problems.

The a.c. (two-phase) servomotor

Preferred at low power, and wherever a synchro error detector has already made the signals a.c. Rugged, light, no brushes. Structurally it is a two-phase induction motor: two stator windings 90° apart in space, excited 90° apart in time, producing a rotating field that drags the rotor round. Two design changes make it a servomotor:

1. **High rotor resistance.** In an ordinary induction motor the reactance-to-resistance ratio is deliberately *large*, which puts peak torque near the operating point — and leaves part of the torque–speed curve *positively* sloped.
2. **Unbalanced excitation.** One winding (the *reference phase*) is fed a fixed voltage; the other (the *control phase*) is fed from the servo amplifier with a variable magnitude at $\pm 90^\circ$. Reverse the sign of the control voltage and the motor reverses. The rotor is a small-diameter squirrel cage or a drag cup, to keep inertia low.

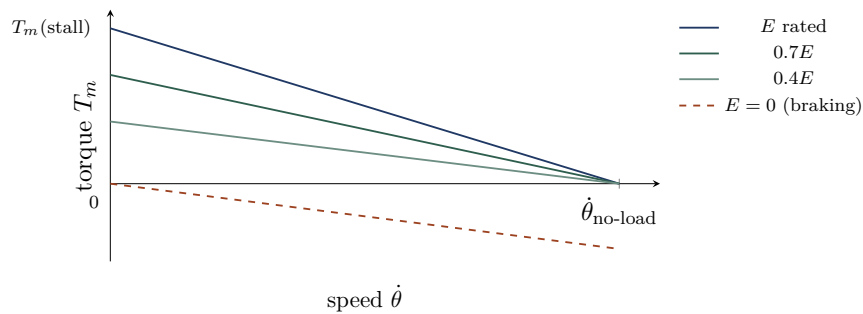


Figure 3: Torque–speed characteristics of a two-phase servomotor at several control-phase voltages, idealised as the straight lines used for linearisation. All slopes are negative — that is the whole point of the high-resistance rotor. At zero control voltage the motor develops a *decelerating* torque, so that line passes through the origin with negative slope. Cf. Nagrath & Gopal Fig. 4.6, p. 86.

Key idea

Why the high rotor resistance matters is a *stability* argument, and it is worth following even without the algebra. The negative slope of the torque–speed curve behaves exactly like extra viscous friction: the faster the motor turns, the more the torque falls off, which damps it. A *positive* slope would behave like *negative* friction — negative damping — and the loop could oscillate on its own. That is why an ordinary induction motor cannot be used as a servomotor, and it is your first sight of a theme that runs to Week 12: stability considerations reach right down into the choice of hardware.

Its transfer function has the same shape as the d.c. motor's: $\theta/E = K_m/[s(\tau_m s + 1)]$.

The stepper motor

A stepper converts a train of pulses into discrete angular steps — **one step per pulse**. It is the actuator of incremental motion control: printers, tape and capstan drives, machine tools, plotters, 3-D printers.

In a *variable-reluctance* stepper the toothed stator stacks are pulse-excited and the toothed rotors are unexcited. Energise a stack and the rotor is pulled to the nearest **minimum-reluctance** position, where the teeth line up. Successive stator stacks are offset by

$$\alpha = \frac{360^\circ}{nT}, \quad n = \text{stacks (= phases)}, \quad T = \text{rotor teeth},$$

and that offset α is the **step angle**. Exciting the phases in the order *abcabc*... steps the rotor one way, *bacbac*... the other — which is why *directional control needs three or more phases*.

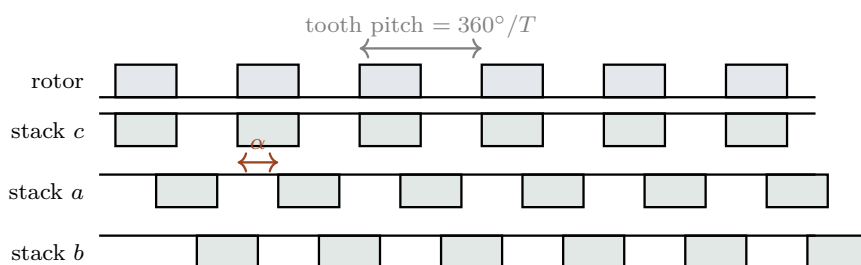


Figure 4: Developed (rolled-out) view of a three-stack variable-reluctance stepper. Rotor teeth are aligned across all stacks; stator stacks are offset successively by $\alpha = 360^\circ/nT$. Energising c , then a , then b pulls the rotor one step α at a time. Cf. Nagrah & Gopal Figs. 4.18 and 4.20, pp. 100–101.

Key idea

The stepper is a **digital actuator**: shaft position is determined entirely by the pulse count. An *open-loop* step servo can therefore reach the accuracy of a closed-loop analogue system with no position sensor at all — the rare case where open loop is the right engineering answer.

Common pitfall

The price is **skipped steps**. If pulses arrive faster than the rotor can settle, or the rotor oscillates too much about the lock position, a step is lost — and since nothing is measured, it is lost *silently and permanently*. High-performance systems close the loop after all, gating the pulse train from a position feedback signal.

Stepper dynamics are genuinely nonlinear and are solved numerically — one reason steppers do not reappear in the analysis weeks.

Full treatment: *main notes §6 (the Taylor-series linearisation of the a.c. servomotor's torque-speed surface, the stall-torque and no-load-speed tests that give K and f , and the reluctance-torque model of the stepper). Nagrah §4.3, p. 84 (a.c. servomotor) and §4.4, p. 99 (stepper).*

4 Sensors: the tachogenerator

A **tachogenerator** produces a voltage proportional to shaft speed:

$$v_t = K_t \dot{\theta}, \quad [K_t] = \text{V rad}^{-1} \text{s}^{-1}.$$

A *d.c.* tachogenerator is a small permanent-magnet generator (commutation ripple is its main limitation); an *a.c. drag-cup* type uses a thin aluminium cup as its rotor, for very low inertia.

It is used two ways: as the **sensor of a speed loop**, where speed is the controlled variable; and as **rate feedback** inside a *position* loop, where the speed signal is subtracted in an inner loop purely to add damping.

Key idea

The second use is the interesting one. Adding rate feedback changes *only* the damping of the closed-loop system; it leaves the steady-state gain alone. So you can calm an oscillatory servo without making it less accurate.

That is your first sight of a **compensator** — a component added not because the plant needs it to function, but because the *loop* needs it to behave. We use it as a design tool in Week 9; FEE3412 studies it properly.

Full treatment: *main notes §7, which works through Nagrath's a.c. position control system and shows the tachometer constant K_t appearing only in the s coefficient of the closed-loop denominator, i.e. only in the damping term. Nagrath §4.3, p. 88 (tachometer) and p. 96 (the full system).*

5 Hydraulic power elements

Where the load is large, hydraulics win: a hydraulic motor is far smaller than an electric one of the same power, and the components are fast and very stiff under load. Against that — leaks, sealing, noise, sluggishness when cold, and inflexible lines. Typical uses: power steering and brakes, ship steering gear, machine tools, aircraft flight controls, excavators.

Output devices split by motion: **rotary** — *hydraulic motors*, which is what the syllabus calls “rotary actuators”; and **translational** — *linear actuators*, i.e. cylinders and rams.

Pump–motor transmission

A **variable-stroke pump** driving a **fixed-stroke motor**, both axial-piston machines whose pistons bear on a wobble plate. Tilt the plate through the *stroke angle* and oil is delivered in proportion to the tilt; reverse the tilt and the motor reverses. One mechanical variable controls everything. A flow balance and a torque balance give $\theta/X = K/[s(\tau s + 1)]$ — the same shape as the electric motors.

Valve control, and the cylinder

Instead of varying the pump stroke, hold the supply pressure constant and throttle the flow with a **spool valve**. Its moving part is far lighter than a pump's stroke mechanism, so time constants shrink and the system becomes much faster. The cost: valve flow is nonlinear — through a sharp-edged orifice it goes as $\sqrt{\Delta p}$ — so the model we use is a *linearisation*, valid only for small movements about neutral.

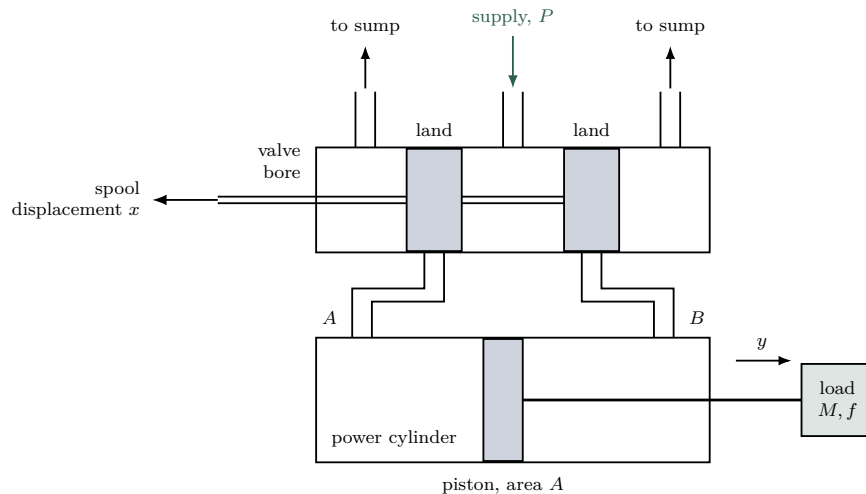


Figure 5: Four-way spool valve controlling a double-acting power cylinder — a hydraulic linear actuator. Spool displacement x admits high-pressure oil to one side of the piston and vents the other to sump; the differential pressure drives the load a distance y . Cf. Nagrath & Gopal Fig. 4.27, p. 111 (the section begins on p. 110).

The cylinder's transfer function is, again, $Y/X = K/[s(\tau s + 1)]$.

Common pitfall

Where the $1/s$ comes from — this is the important sentence in the whole document. It is not an approximation and it is not inertia. Flow into a cylinder sets the piston's *velocity*; position is therefore the accumulation of that velocity. **Any actuator whose input commands a rate** — valve opening $\rightarrow$ velocity, armature voltage $\rightarrow$ speed, pump stroke $\rightarrow$ motor speed — carries a free integrator when the output you care about is position.

Full treatment: *main notes §8 (the continuity and torque balances, the leakage and compressibility flow terms, and the linearised valve relation $q = K_1x - K_2p$). Nagrath §4.5: pumps and motors p. 104, valves p. 108, linear actuator p. 110.*

6 Pneumatic power elements

Air instead of oil: non-inflammable, intrinsically safe, and with a viscosity that is negligible and stable with temperature. But air is **compressible**, so pneumatic systems carry substantial compressibility flow and are characterised by **longer time delays** — which is why they dominate in *process* control, where the plant is slow anyway, and are rare where speed matters.

Bellows

A thin corrugated chamber that behaves as a spring: pressure in, displacement out, with no dynamics — a *pure gain* A/K .

Flapper–nozzle valve

The key sensing element. Air at constant supply pressure passes through a fixed orifice and out of a nozzle; a pivoted flapper sits close to the nozzle. Move the flapper nearer and the restriction rises, so the back pressure climbs towards the supply; move it away and the pressure falls towards ambient. A very small movement produces a very large pressure change — a high-gain displacement-to-pressure transducer.

Pneumatic relay

Keeping the flapper movement small enough to stay linear also keeps the output pressure swing small, so a pneumatic *power amplifier* is added in cascade. A ball seats to either connect the output to supply or vent it to atmosphere. The relay also inverts the sign.

Diaphragm actuator

Provides the translational output most pneumatic systems need: pressure acts on a diaphragm, which drives a stem against a return spring.

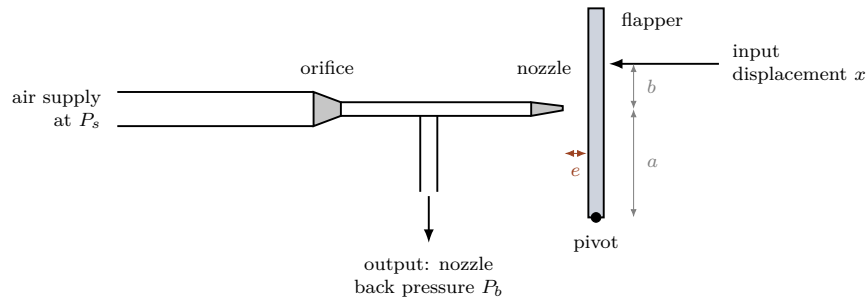


Figure 6: Flapper–nozzle valve. Small movements of the flapper produce large changes in back pressure — a high-gain displacement-to-pressure transducer. Cf. Nagraath & Gopal Fig. 4.34, p. 117.

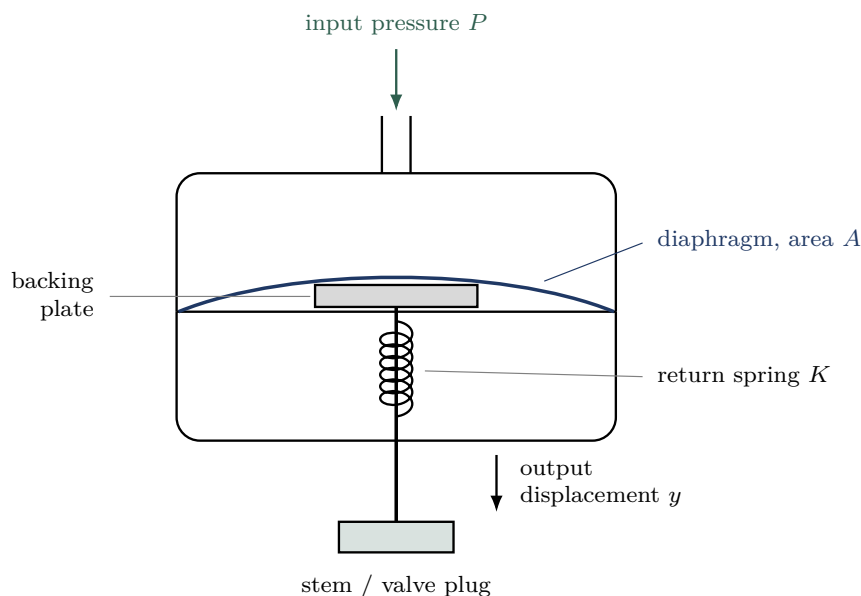


Figure 7: Pneumatic diaphragm actuator. Unlike the hydraulic cylinder, the spring gives it a definite equilibrium position for each pressure — so its transfer function is second order with *no* free integrator. Cf. Nagraath & Gopal Fig. 4.37, p. 119.

Common pitfall

Compare the two actuators. The hydraulic cylinder has a free $1/s$; the spring-loaded pneumatic diaphragm does **not** — it is $A/(Ms^2 + fs + K)$, an ordinary second-order lag. The difference is the **spring**. With a spring, a constant pressure gives a constant *position*, because the spring force balances the applied force at a definite displacement. Without one, a constant flow gives a constant *velocity*. Read the physics, not the picture.

Full treatment: *main notes §9 (the force balances giving each transfer function, and the lever-arm factor $a/(a+b)$ in the flapper gain). Nagrath §4.6: bellows p. 116, flapper valve p. 117, relay p. 118, actuator p. 119.*

7 Choosing between them

	Electric	Hydraulic	Pneumatic
Working medium	electric current	incompressible oil	compressible air
Power/weight	moderate	very high	low
Speed of response	fast	fastest under load	slow (compressibility)
Stiffness under load	moderate	very high	low
Typical use	instruments, drives	servos, machine tools, steering, aircraft controls, presses	process control valves
Main drawback	limited torque density	leaks, sealing, contamination, noise	long time delays, low stiffness
Safety	sparks	fire risk from oil mist	non-inflammable, intrinsically safe

Key idea

The table is not a list to memorise; it is three trade-offs to reason with. **Power density** pushes you towards hydraulics, **safety and cost** towards pneumatics, **precision and convenience** towards electrics. Almost every real selection is one of those three arguments winning.

8 One shape, over and over

Collect what the survey produced. Note that the *physics* of these devices have almost nothing in common — current and flux, oil under pressure, compressed air — and yet:

Device	Shape	Why
d.c. servomotor	$\frac{K}{s(\tau s + 1)}$	voltage commands <i>speed</i> ; inertia and friction resist
a.c. servomotor	$\frac{K}{s(\tau s + 1)}$	control voltage commands <i>speed</i> ; same resistance
Hydraulic transmission	$\frac{K}{s(\tau s + 1)}$	stroke angle commands <i>motor speed</i>
Hydraulic cylinder	$\frac{K}{s(\tau s + 1)}$	valve opening commands <i>piston velocity</i>
Potentiometer, synchro, tachogenerator, bellows	pure gain	they <i>report</i> a quantity; nothing accumulates
Pneumatic diaphragm actuator	$\frac{A}{Ms^2 + fs + K}$	the spring fixes a <i>position</i> for each pressure, so nothing accumulates

Key idea

Almost every **positional actuator** in control engineering, whatever its physics, has the transfer function

$$G(s) = \frac{K}{s(\tau s + 1)}.$$

The $1/s$ is there because **the input commands a rate** and the output of interest is the accumulation of that rate. The $1/(\tau s + 1)$ is there because **inertia and friction resist** that rate. Sensors and error detectors, by contrast, are usually **pure gains** — they report, they do not accumulate.

The exception proves the rule: put a spring on the output and the accumulation stops, and with it the integrator.

This is not a curiosity to note and forget. It is the reason a single body of mathematics carries the whole unit, and it is why the weeks ahead look the way they do:

- Put $K/[s(\tau s + 1)]$ in a unity-feedback loop and you get a **second-order system** — *Week 5*, and its ζ and ω_n .
- The free integrator makes the open loop **type 1**: it tracks a step with zero steady-state error, a ramp with a finite one — *Week 6*.
- Its pole-zero pattern — one pole at the origin, one on the negative real axis — gives the most common **root locus** you will ever sketch — *Week 8*.
- Its **Bode plot** is a -20 dB/decade line plus a single corner — *Week 10*.

Key idea

If you take one thing from Part B, take this: **you have already met the plant you will study for the next eleven weeks**. The hardware survey was not a digression before the real course — it was the course, in physical form.

Where the mathematics lives

Everything stated here without proof is derived in `week-01-notes.pdf` — the *Full treatment* line under each section above names the exact place.

The primary hardware source throughout is **Nagrath & Gopal, 2nd edition, Chapter 4** (pp. 84–121), with the potentiometer servomechanism at §5.4, p. 137. Beware: Nagrath's section numbering is *not* stable across editions, so check any page number against the copy you actually hold.

Looking ahead

Part A gave you the vocabulary; this document gave you the hardware. From Week 2 the subject becomes mathematical, and it does so by taking exactly these devices and deriving from first principles the transfer functions quoted here without proof — starting with the armature-controlled d.c. servomotor driving a load through a gear train.

Two boundaries worth restating. *Components are survey depth in FEE3411*: you need to describe working principles, not design a servomotor or select a valve. And *FEE3411 owns analysis while FEE3412 owns design* — FEE3412 contains a full hardware block and will treat all of this properly.